

Statistics Programs Record

Experiment 1: Finding Mean Using Python

Aim:

To find the mean of the given data.

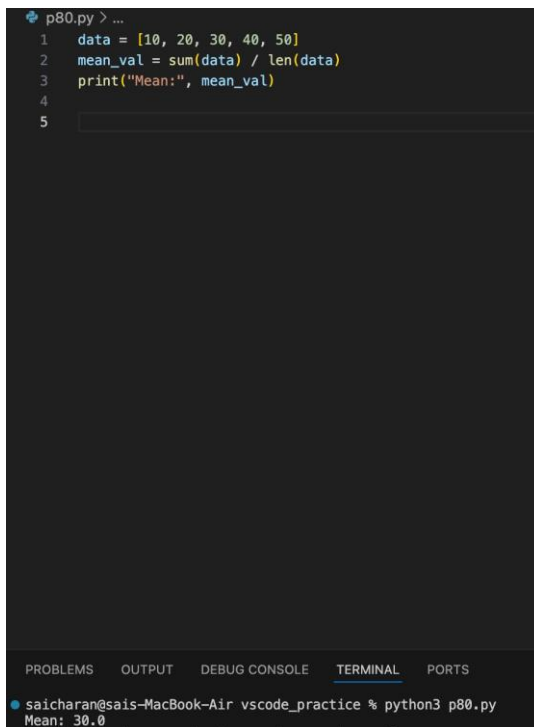
Algorithm:

1. Start.
2. Create a list of numbers.
3. Find the sum of all elements.
4. Divide the sum by the total number of elements.
5. Display the mean.
6. Stop.

Code:

```
data = [10, 20, 30, 40, 50]
mean_val = sum(data) / len(data)
print("Mean:", mean_val)
```

Output:



The screenshot shows a VS Code editor with a file named 'p80.py' containing the following Python code:

```
1 data = [10, 20, 30, 40, 50]
2 mean_val = sum(data) / len(data)
3 print("Mean:", mean_val)
4
5
```

Below the editor, the 'TERMINAL' panel shows the command 'python3 p80.py' being executed, resulting in the output 'Mean: 30.0'.

Result:

The mean of the given data is calculated successfully.

Experiment 2: Finding Median Using Python

Aim:

To find the median of the given data.

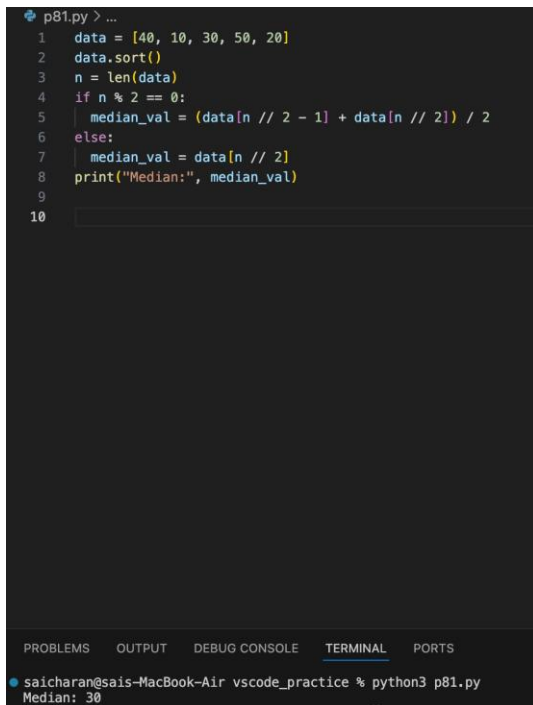
Algorithm:

1. Start.
2. Create a list.
3. Sort the list.
4. Find the middle value.
5. Display the median.
6. Stop.

Code:

```
data = [40, 10, 30, 50, 20]
data.sort() n = len(data) if n % 2 == 0:    median_val =
(data[n // 2 - 1] + data[n // 2]) / 2 else:
    median_val = data[n // 2]
print("Median:", median_val)
```

Output:



```
p81.py > ...
1  data = [40, 10, 30, 50, 20]
2  data.sort()
3  n = len(data)
4  if n % 2 == 0:
5      median_val = (data[n // 2 - 1] + data[n // 2]) / 2
6  else:
7      median_val = data[n // 2]
8  print("Median:", median_val)
9
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

saicharan@sais-MacBook-Air vscode_practice % python3 p81.py
Median: 30

Result: The median is calculated successfully.

Experiment 3: Finding Mode Using Python

Aim:

To find the mode of the given data.

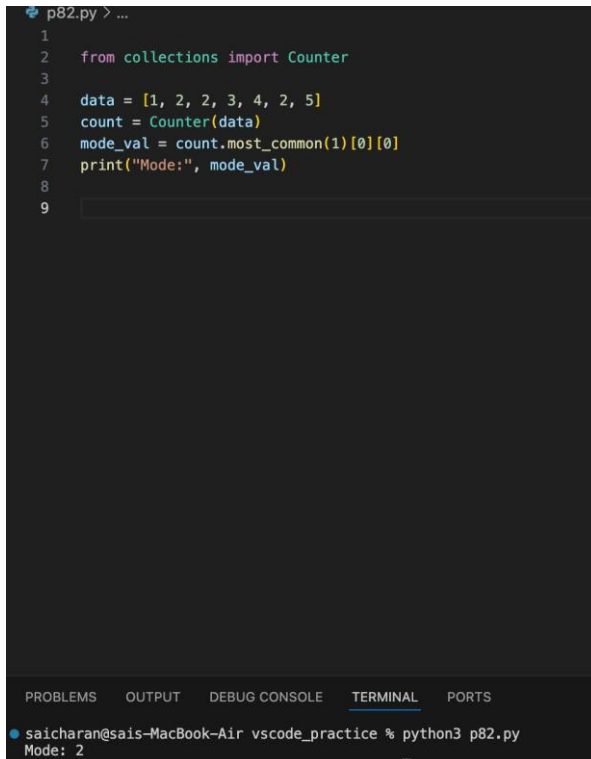
Algorithm:

1. Start.
2. Import Counter.
3. Create a list.
4. Count the occurrences.
5. Find the most common value.
6. Display the mode.
7. Stop.

Code:

```
from collections import Counter
data = [1, 2, 2, 3, 4, 2, 5] count =
Counter(data) mode_val =
count.most_common(1)[0][0]
print("Mode:", mode_val)
```

Output:



The screenshot shows a VS Code editor with a file named p82.py. The code in the editor is as follows:

```
1
2 from collections import Counter
3
4 data = [1, 2, 2, 3, 4, 2, 5]
5 count = Counter(data)
6 mode_val = count.most_common(1)[0][0]
7 print("Mode:", mode_val)
8
9
```

Below the editor, the TERMINAL panel shows the command and output:

```
saicharan@sais-MacBook-Air vscode_practice % python3 p82.py
Mode: 2
```

Result: The mode is calculated successfully.

Experiment 4: Finding Range Using Python

Aim:

To find the range of the given data.

Algorithm:

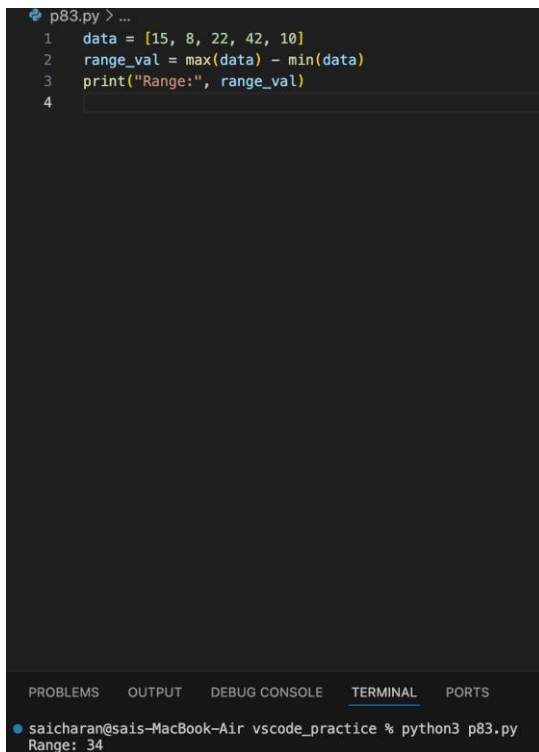
1. Start.
2. Create a list.
3. Find the maximum value.
4. Find the minimum value.
5. Subtract the minimum from the maximum.
6. Display the range.
7. Stop.

Code:

Code:

```
data = [15, 8, 22, 42, 10] range_val =  
max(data) - min(data) print("Range:",  
range_val)
```

Output:



```
p83.py > ...  
1 data = [15, 8, 22, 42, 10]  
2 range_val = max(data) - min(data)  
3 print("Range:", range_val)  
4  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
saicharan@sais-MacBook-Air vscode_practice % python3 p83.py  
Range: 34
```

Result: The range is calculated successfully.

Experiment 5: Finding Variance Using Python

Aim:

To find the population variance.

Algorithm:

1. Start.
2. Create a list.
3. Calculate the mean.
4. Calculate squared differences.
5. Find the average.
6. Display the variance.
7. Stop.

Code:

```
data = [2, 4, 4, 4, 5, 5, 7, 9] mean_val =  
sum(data) / len(data)  
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)  
print("Variance:", variance_val)
```

Output:



The screenshot shows a VS Code editor with a file named `p84.py`. The code in the editor is as follows:

```
1 data = [2, 4, 4, 4, 5, 5, 7, 9]  
2 mean_val = sum(data) / len(data)  
3 variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)  
4 print("Variance:", variance_val)  
5  
6
```

At the bottom, the `TERMINAL` panel shows the command `python3 p84.py` being executed, resulting in the output:

```
Variance: 4.0
```

Result:

The variance is calculated successfully.

Experiment 6: Finding Standard Deviation Using Python**Aim:**

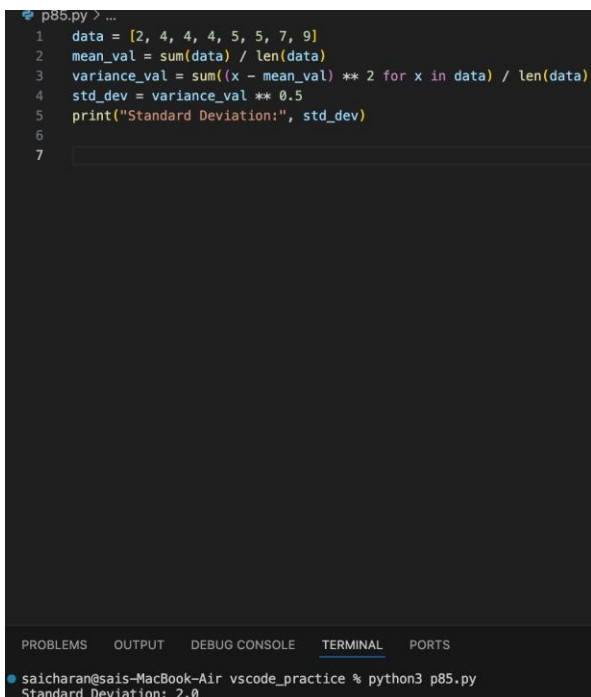
To find the standard deviation.

Algorithm:

1. Start.
2. Create a list.
3. Calculate the mean.
4. Find the variance.
5. Take the square root.
6. Display the standard deviation.
7. Stop.

Code:

```
data = [2, 4, 4, 4, 5, 5, 7, 9] mean_val =  
sum(data) / len(data)  
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)  
std_dev = variance_val ** 0.5  
print("Standard Deviation:", std_dev)
```

Output:

```
p85.py > ...  
1 data = [2, 4, 4, 4, 5, 5, 7, 9]  
2 mean_val = sum(data) / len(data)  
3 variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)  
4 std_dev = variance_val ** 0.5  
5 print("Standard Deviation:", std_dev)  
6  
7  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
saicharan@sais-MacBook-Air vscode_practice % python3 p85.py  
Standard Deviation: 2.0
```

Result:

The standard deviation is calculated successfully.

Experiment 7: Using statistics Module**Aim:**

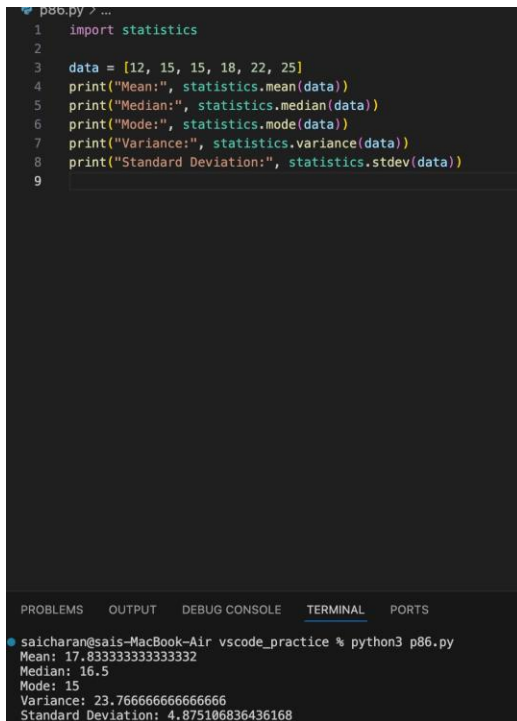
To calculate statistical measures using the statistics module.

Algorithm:

1. Start.
2. Import the statistics module.
3. Create a list.
4. Calculate statistical measures.
5. Display the output.
6. Stop.

Code:

```
import statistics
data = [12, 15, 15, 18, 22, 25] print("Mean:",
statistics.mean(data)) print("Median:",
statistics.median(data)) print("Mode:",
statistics.mode(data)) print("Variance:",
statistics.variance(data)) print("Standard Deviation:",
statistics.stdev(data))
```

Output:

The screenshot shows a code editor with a Python script named p86.py. The script imports the statistics module, creates a list 'data' with values [12, 15, 15, 18, 22, 25], and prints the mean, median, mode, variance, and standard deviation. Below the code editor, a terminal window displays the output of the script, which matches the values calculated in the code.

```
p86.py > ...
1  import statistics
2
3  data = [12, 15, 15, 18, 22, 25]
4  print("Mean:", statistics.mean(data))
5  print("Median:", statistics.median(data))
6  print("Mode:", statistics.mode(data))
7  print("Variance:", statistics.variance(data))
8  print("Standard Deviation:", statistics.stdev(data))
9

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
saicharan@sais-MacBook-Air vscode_practice % python3 p86.py
Mean: 17.833333333333332
Median: 16.5
Mode: 15
Variance: 23.766666666666666
Standard Deviation: 4.875106836436168
```

Result:

Statistical measures are calculated successfully.

Experiment 8: Using NumPy Library**Aim:**

To calculate statistical measures using NumPy.

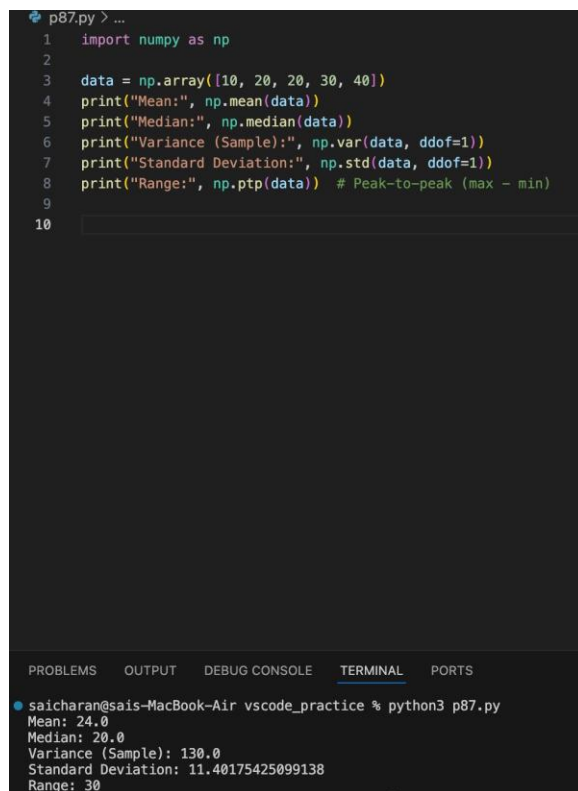
Algorithm:

1. Start.
2. Import NumPy.
3. Create an array.
4. Apply NumPy functions.
5. Display the results.
6. Stop.

Code:

```
import numpy as np
```

```
data = np.array([10, 20, 20, 30, 40]) print("Mean:",  
np.mean(data)) print("Median:", np.median(data))  
print("Variance (Sample):", np.var(data, ddof=1))  
print("Standard Deviation:", np.std(data, ddof=1))  
print("Range:", np.ptp(data))
```

Output:

```
p87.py > ...  
1 import numpy as np  
2  
3 data = np.array([10, 20, 20, 30, 40])  
4 print("Mean:", np.mean(data))  
5 print("Median:", np.median(data))  
6 print("Variance (Sample):", np.var(data, ddof=1))  
7 print("Standard Deviation:", np.std(data, ddof=1))  
8 print("Range:", np.ptp(data)) # Peak-to-peak (max - min)  
9  
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● saicharan@sais-MacBook-Air vscode_practice % python3 p87.py  
Mean: 24.0  
Median: 20.0  
Variance (Sample): 130.0  
Standard Deviation: 11.40175425099138  
Range: 30
```


Result:

Statistical measures are calculated successfully using NumPy.

Experiment 9: Using SciPy for Mode**Aim:**

To find the mode using SciPy.

Algorithm:

1. Start.
2. Import SciPy.
3. Create a list.
4. Find the mode.
5. Display the mode and count.
6. Stop.

Code:

```
from scipy import stats
data = [5, 2, 5, 3, 5, 4] mode_result =
stats.mode(data, keepdims=True) print("Mode:",
mode_result.mode[0]) print("Count:",
mode_result.count[0])
```

Output:The image shows a screenshot of a VS Code editor window. The editor has a dark theme. At the top, a file named 'p88.py' is open. The code in the editor is as follows:

```
1 from scipy import stats
2
3 data = [5, 2, 5, 3, 5, 4]
4 mode_result = stats.mode(data, keepdims=True)
5 print("Mode:", mode_result.mode[0])
6 print("Count:", mode_result.count[0])
7
8
```

Below the editor, there is a panel with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is selected, showing the output of running the script:

```
saicharan@sais-MacBook-Air: vscode_practice % python3 p88.py
Mode: 5
Count: 3
```

Result: Mode is calculated successfully using SciPy.

Experiment 10: Using Pandas for Complete Summary

Aim:

To calculate summary statistics using Pandas.

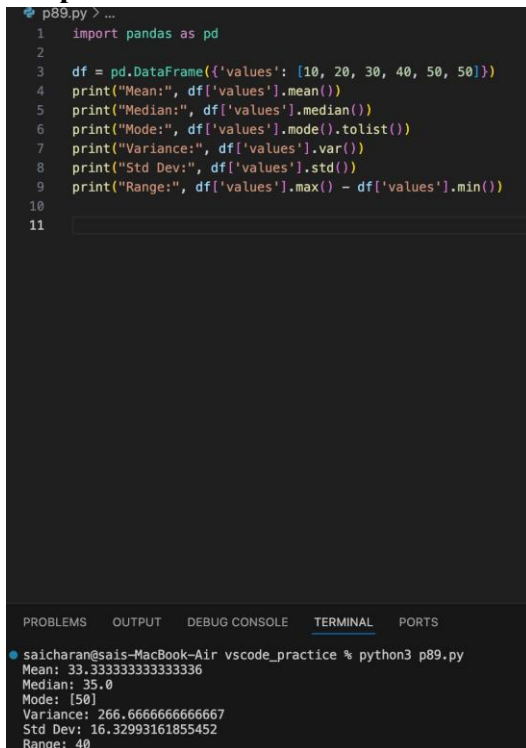
Algorithm:

1. Start.
2. Import pandas.
3. Create a DataFrame.
4. Calculate statistical measures.
5. Display the results.
6. Stop.

Code:

```
import pandas as pd
df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})
print("Mean:", df['values'].mean()) print("Median:",
df['values'].median()) print("Mode:",
df['values'].mode().tolist()) print("Variance:",
df['values'].var()) print("Std Dev:", df['values'].std())
print("Range:", df['values'].max() - df['values'].min())
```

Output:



```
p89.py > ...
1  import pandas as pd
2
3  df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})
4  print("Mean:", df['values'].mean())
5  print("Median:", df['values'].median())
6  print("Mode:", df['values'].mode().tolist())
7  print("Variance:", df['values'].var())
8  print("Std Dev:", df['values'].std())
9  print("Range:", df['values'].max() - df['values'].min())
10
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
saicharan@sais-MacBook-Air vscode_practice % python3 p89.py
Mean: 33.33333333333333
Median: 35.0
Mode: [50]
Variance: 266.6666666666667
Std Dev: 16.32993161855452
Range: 40
```

Result:

Summary statistics are calculated successfully using Pandas.